

Discovery Findings & Updated Project Scope

Duke Energy

December 2025



Executive Summary: Discovery Insights

In-Depth Discovery

We conducted comprehensive discovery sessions with your team to refine requirements for the custom application. These sessions provided critical clarity on workflows, user needs, and system integration points.

Three-Dimensional Impact

Discovery reshaped project scope across three key dimensions: strategic additions of new functionality, thoughtful modifications to original features, and selective removals of non-essential elements.

Net Scope Increase

While some original scope was simplified or removed, substantial new functionality was identified as business-critical. The overall project scope has increased from original estimates.

Key Drivers of Scope Growth

- Enhanced understanding of system complexity and integration requirements
- Need for "stopgap" functionality bridging current MVP to future full integration
- Addition of entirely new feature sets not fully defined in original RFP
- Expanded user role complexity (7 roles vs. 3 originally planned)

Strategic Value

The updated scope better reflects actual business needs and provides a more realistic, executable path to a functional system that serves your operations effectively.

Pricing reflects full-scope comparison and includes contingency for known unknowns, particularly around third-party integration timing and API availability.

Discovery Process: From RFP to Reality

- 1** — Original RFP Scope
Initial requirements provided a solid foundation but lacked detail on workflow nuances, integration dependencies, and operational realities.
- 2** — Discovery Sessions
Deep-dive workshops with stakeholders revealed critical insights about FSM integration, Dynamics CRM dependencies, and diverse user role requirements.
- 3** — Refined Understanding
Gained comprehensive clarity on workflows, third-party tool limitations, and the need for bridge functionality during transition periods.
- 4** — Updated Roadmap
Developed realistic phasing approach with accurate scope definition, timeline expectations, and integration sequencing.

The Benefit: A more accurate roadmap with realistic phasing that accounts for technical dependencies, operational constraints, and business priorities. This foundation ensures successful delivery rather than setting unrealistic expectations.



Scope Changes: What Was Removed or Simplified

Removed Entirely

- **Contractor Marketplace:** Deferred to future phase
- **In-App Rating Display:** Moved out of MVP scope
- **Calendar Integration:** Eliminated for contractors
- **DIY Content Section:** Replaced with better alternative

Simplified Features

- **Automation Workflows:** Phase 1 relies more on CSRs/schedulers with manual tools
- **Contractor Matching:** Streamlined algorithm approach
- **System Complexity:** Reduced by leveraging existing Dynamics CRM and FSM2

Cost Impact

These removals and simplifications reduce costs in specific areas, creating budget capacity for higher-priority functionality identified during discovery.

Strategic trade-offs ensure resources focus on features delivering immediate business value.



Scope Changes: What Was Modified and Enhanced

Claims Management

Impact: Net Increase

Significantly enhanced with better workflow understanding. Added critical dimension of ad hoc customers vs. existing customers (major addition). Simpler workflow structure but more comprehensive feature set overall.

Home Inventory System

Impact: Net Increase

More advanced functionality with greater detail level and data capture. While some features were removed, net additions are significantly larger, providing homeowners with comprehensive inventory tracking.

Service Booking & Contractor Management

Impact: Net Decrease

Removed: Full contractor self-service portal

Added: CSR browse/lookup capabilities for account management

Rationale: Avoid CSRs juggling four different systems; phone-only for contractors with robust internal tools

DIY Content Replacement

Impact: Net 0

Removed: Static Do-It-Yourself content library

Added: Dynamic homeowner recommendations system with automated notifications

Net Impact: Increase, as notifications require backend infrastructure and smart logic vs. static content display

Scope Changes: Entirely New Features Added



Ad Hoc vs. Member Customer Separation

Complete new workflow and user management system distinguishing between one-time customers and membership holders. Requires dual pathways for claims, service booking, and account management.



Contractor Database & Matrix

New internal system managing contractors before FSM tool implementation. Tracks service areas, geographical coverage, preferred vs. backup status, and availability. Critical bridge functionality.



Enhanced Analytics & Tracking

Comprehensive interaction tracking across all touchpoints. Provides visibility into customer behavior, service patterns, and operational efficiency metrics.



Gamification System

"Home Score" rating system for homeowners encouraging engagement and preventive maintenance. Includes point accumulation, achievement badges, and progress tracking.



Maintenance Reminders

Complete new feature set including scheduling, automated notifications, tracking, and completion verification. Proactive homeowner engagement tool.



Expanded User Roles

Original scope: ~3 user roles. Updated scope: ~7 different roles with distinct permissions, workflows, and interface requirements. Significantly increased complexity.

The "Bridge Gap" Challenge



The Core Complexity Driver

Your organization needs a functional MVP now, but full third-party tools (FSM API, complete Dynamics CRM integration) won't be available until later phases. This timing gap creates a critical challenge.

Without FSM API and complete Dynamics CRM integration available at launch, we need functional workarounds. Examples include: manual contractor assignment instead of automated routing, admin-managed status updates instead of real-time sync, and static availability calculations instead of dynamic scheduling.

		
The Challenge Operations cannot wait for full third-party integrations. Business needs immediate functionality to serve customers and manage contractors effectively. FSM/API availability: 6-12 months post-launch Cost of waiting: \$2M+ in lost revenue	Our Solution Build "stopgap" functionality: manual browse/management screens for CSRs, temporary contractor database, and operational tools enabling work during transition. Build abstraction layers (\$25,500 investment) that allow backend swaps without UI changes Manual workflows with admin tools bridge the gap	Future Phase Some stopgap features will be replaced or removed once third-party APIs are fully live and integrated, transitioning to automated workflows. Expected rework: \$46K (reduced from \$96K through smart architecture) Gradual cutover minimizes disruption

 **Cost Impact:** Bridge functionality adds \$177,750 to Phase 1 (\$152,250 temporary features + \$25,500 abstraction layers).

Detailed Breakdown: Risk of Obsolescence in Temporary Features

Components That Will Be Replaced When FSM Tool Arrives

SERVICE REQUEST STATUS MANAGEMENT

Current Approach: Manual admin dashboard for status updates, simplified workflow engine, admin notification system

When FSM Arrives: Status updates become automated, real-time sync replaces manual updates, admin role shifts to exception handling

Obsolescence: 70% waste

CONTRACTOR AVAILABILITY & LEAD TIME LOGIC

Current Approach: Static availability configuration, fixed lead time buffers, manual capacity tracking

When FSM Arrives: Real-time calendar sync, dynamic scheduling with instant confirmation, FSM manages all capacity

Obsolescence: 75% waste

CONTRACTOR ASSIGNMENT ALGORITHM

Current Approach: Simple trade + zip code matching, primary/secondary designation, manual override interface

When FSM Arrives: Intelligent routing using proximity, ratings, capacity, and real-time availability

Obsolescence: 50% waste

APPOINTMENT CONFIRMATION WORKFLOW

Current Approach: Admin emails/calls contractors, manual response tracking, backup escalation process

When FSM Arrives: Instant confirmation via contractor app, automated escalation, real-time customer updates

Obsolescence: 75% waste

SERVICE REQUEST TRACKING

Current Approach: 5 basic statuses (Pending → Confirmed → Completed), limited customer visibility

When FSM Arrives: Rich status updates (Dispatched, En Route, On-Site, In Progress), GPS tracking, "pizza tracker" experience

Obsolescence: 50% waste

Understanding the Temporary Build Challenge

Features at Risk When FSM/APIs Arrive

Eight major components built for Phase 1 will require significant rework once third-party integrations are available.



Key Metrics:

- Significant temporary development effort
- Investment in stopgap solutions
- Majority of effort will require rework
- Portion of effort is reusable

HIGH-RISK OBSOLESCENCE (67% waste)

- Manual status management
- Static availability logic
- Simplified assignment algorithm
- Manual confirmation workflow
- Limited status tracking

MEDIUM-RISK OBSOLESCENCE (71% waste)

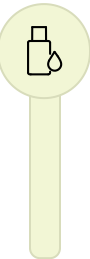
- API fallback processes
- Contractor data management

LOW-RISK OBSOLESCENCE (25% waste)

- Notification system (will enhance, not replace)

 **Significant portion of Phase 1 investment is at risk.** With mitigation strategies, expected waste can be substantially reduced.

Revised Project Approach



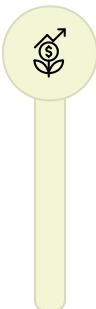
1: Core (Must-Have)

Absolute essentials for system to function. Basic claims processing, service request intake, home inventory foundation, user authentication, and core database structure. Minimal viable operations.



2: Stopgap/Bridge

Functional system with temporary solutions for missing third-party integrations. CSR management tools, contractor browse/lookup, manual scheduling support, and operational dashboards. Enables full business operations.



3: Additions & Enhancements

Advanced features including gamification, enhanced analytics, automated recommendations, plus replacement of stopgap solutions with full API integrations and automated workflows. Optimized efficiency.

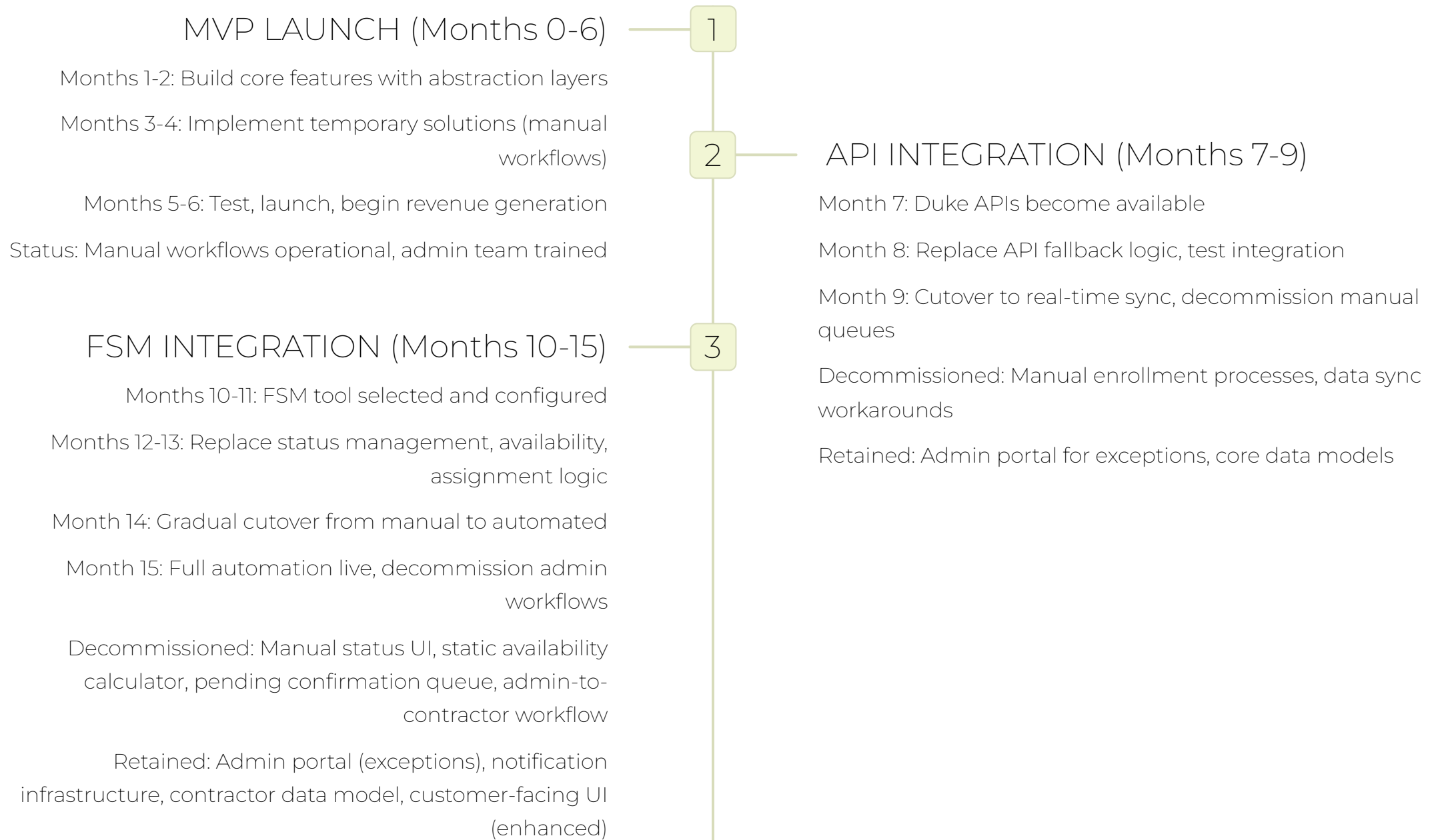
This approach balances immediate operational needs with long-term system optimization, ensuring you can begin serving customers while building toward the complete vision.

Project Roadmap Business Planning



Integration Timeline & Transition Plan

Approach to Minimize Disruption



Rework investment required across transition phases

Significant revenue generation during transition period

Mitigation Strategy: Design for Abstraction

Reducing Future Rework Through Smart Architecture

The Problem Without Mitigation

- Building temporary features directly = 63% obsolescence
- Substantial waste expected
- Difficult FSM/API integration later
- Significant rework required

The Solution: Abstraction Layers

- Build temporary features behind abstraction interfaces
- Swap implementations without changing UI
- Gradual cutover from manual to automated
- Waste reduced by approximately 50%



Service Request Abstraction Layer

Benefit: Reduce status management waste from 70% to 30%



Contractor Data Service Interface

Benefit: Reduce assignment waste from 50% to 20%



Availability Provider Pattern

Benefit: Reduce availability waste from 75% to 30%



API Gateway Pattern (Duke APIs)

Benefit: Reduce API fallback waste from 75% to 25%



Event-Driven Architecture Prep

Benefit: Enable easier FSM integration

 **Mitigation Investment:** Additional architectural effort

Net Benefit: Significant reduction in future rework

ROI: Nearly 2x return on mitigation investment

Cost Impact Summary

Budget Evolution

Original Scope Estimate: Based on initial RFP assumptions

Updated Scope Estimate: Reflects discovery insights and bridge functionality requirements

The increase includes \$177,750 for temporary features and abstraction layers—a strategic investment enabling 6-12 months earlier launch and \$2M+ revenue capture during the FSM/API transition period.

Highest Investment Areas

1. Claims Management System (+\$70K from enhanced workflows)
2. Contractor Management & Matching (includes temporary assignment logic)
3. Bridge/Stopgap Functionality (\$152,250 temporary features)
4. User Role Management (7 roles vs. 3 originally planned)
5. Analytics & Tracking Infrastructure
6. Notifications & Recommendations Engine
7. Abstraction Layer Architecture (\$25,500 mitigation investment)



Claims Enhancements

~\$70K impact from comprehensive workflow improvements and ad hoc customer support

Includes dual pathways for members vs. one-time customers



Bridge Functionality

\$152,250 temporary features + \$25,500 abstraction = \$177,750

Enables operations during 6-12 month FSM/API development phase

Expected \$46K rework cost (reduced from \$96K through mitigation)



ROI Analysis

Temporary investment: \$177,750

Expected waste: \$46,112 (6% of project with mitigation)

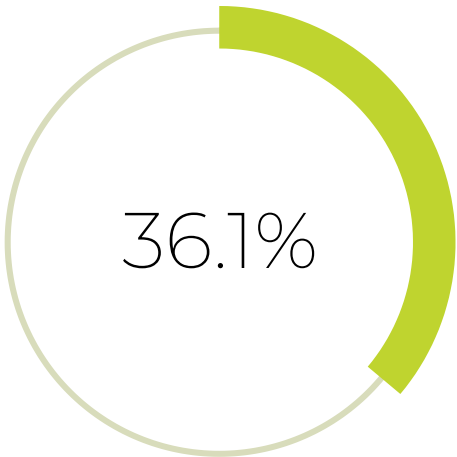
Revenue during transition: \$2M+ in Ad Hoc revenue

Net ROI: 11.7x return

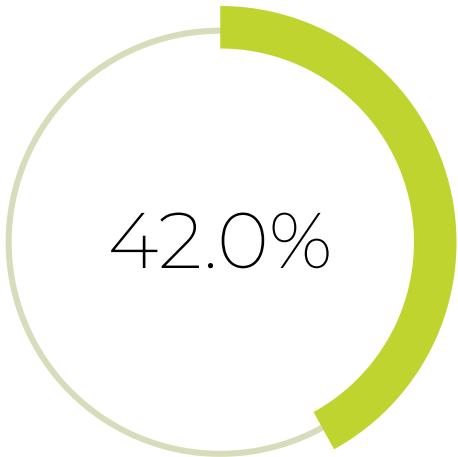
Business Case: The \$177,750 investment in temporary features and abstraction layers enables launch 6-12 months earlier, generating \$2M+ in revenue during the transition period. Even accounting for \$46K in eventual rework, this represents an 11.7x return on investment. Waiting for full FSM/API availability would cost \$2M+ in lost revenue and competitive positioning.

Project Budget Overview

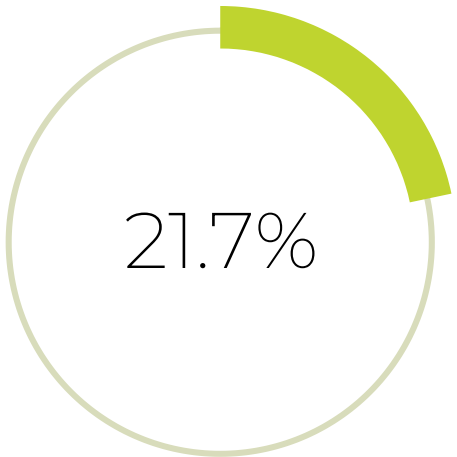
Investment Breakdown by Component



CX Interface Development
Mobile Responsive + Native Apps
\$285,225



Backend Development
Laravel + API Infrastructure
\$332,083



DevOps & Security
Deployment, Infrastructure & Compliance
\$171,292

Total Development Investment

\$788,600

Ongoing Costs

- Hosting: \$800/month (includes beta + production environments, scales with user growth)
- Support & Maintenance: \$6,500/month (optional but recommended)

Key Notes: Budget reflects comprehensive discovery findings including bridge functionality, enhanced claims workflows, dual service pathways (HPP + ad-hoc), and 7-role admin architecture. Hosting costs will scale with user growth from 200K to 2M users over 3 years.

CX Interface Development Breakdown

Customer-Facing Application Components (\$285,225)

AUTHENTICATION & USER MANAGEMENT - User registration & onboarding flow - Multi-property support - Profile creation separate from utility accounts - Session management & guest browsing	CUSTOMER PROFILE & DASHBOARD - Multi-property management interface - Profile management (personal info, preferences) - Communication preferences - Property selection/switching	HOME PROTECTION PLAN (HPP) MANAGEMENT - View active HPP plans linked to profile - New plan enrollment workflow - Upgrade/downgrade existing plans - Self-service cancellation flow - Integration with Duke API for plan validation
HOME INVENTORY/PROFILE BUILDING - Manual appliance/system entry - Barcode scanning for auto-populate - Pre-fill home characteristics from address - Photo upload for inventory items - Warranty tracking & maintenance schedule generation - Recall checking integration (CPSC database)	SERVICE BOOKING - HPP COVERED SERVICES - Symptom-based intake form - Coverage check logic - Contractor assignment display - Date/time selection with static availability buffers - Booking confirmation flow - Emergency detection logic	SERVICE BOOKING - AD-HOC SERVICES - Browse/search service catalog - Service detail pages with transparent pricing - Flat-rate & variable pricing display - Payment collection messaging - Price display by region
MOBILE NATIVE APP DEVELOPMENT - iOS native app (Swift/SwiftUI or React Native) - Android native app (Kotlin or React Native) - App store deployment setup - Push notification configuration - Camera integration for barcode scanning - Deep linking & offline capability	COMMUNICATION & NOTIFICATIONS - Push notification setup (iOS/Android) - SMS & email notification integration - In-app notification center - Multi-channel delivery logic	MAINTENANCE REMINDERS & GAMIFICATION - Inventory-based reminders display - Profile completion score indicator - Loyalty points tracking system - Home health scorecard display - Badge system (achievements)

 Includes responsive web interface, mobile-first design, accessibility compliance (WCAG 2.1), and complete UI/UX design system implementation.

Backend Development Breakdown

API & Business Logic Infrastructure (\$332,083)

CORE BACKEND ARCHITECTURE - Laravel application setup - Database schema design & implementation - RESTful API architecture - Authentication & authorization (JWT/OAuth) - Multi-tenant architecture support - API documentation (Swagger/OpenAPI)	HPP PLAN MANAGEMENT BACKEND - HPP plan data model - Duke Enterprise API integration (customer validation & HPP plans) - Plan enrollment processing - Plan upgrade/downgrade logic - Subscription status sync (Commerce/Dynamics) - Multi-plan support per property	HOME INVENTORY MANAGEMENT - Inventory data model & CRUD API endpoints - Barcode lookup API integration - Public data pre-fill API (address → home characteristics) - Photo upload & storage (S3/cloud) - Maintenance schedule generation logic - CPSC recall checking integration - Warranty tracking logic
CONTRACTOR MATCHING ALGORITHM - Contractor data model (complex multi-trade support) - Trade + Zip code matching logic - Primary/secondary designation handling - Service area configuration (zip code lists) - Lead time calculation logic - Availability buffer implementation - Override/exception handling	SERVICE REQUEST MANAGEMENT - HPP - Service request data model - Coverage check logic (HPP eligibility) - Contractor assignment automation - Status management workflow - Manual status update capability for admin - Duke CRM API integration (if available) - Manual fallback queue processing - Emergency detection logic	AD-HOC SERVICE CATALOG MANAGEMENT - Ad-hoc service catalog data model (NEW - doesn't exist) - Service CRUD APIs - Pricing management (flat-rate, variable, quote) - Geographic pricing support - Scope of work definition - Service availability by region - Contractor-service association
COMMUNICATION & NOTIFICATION SYSTEM - Multi-channel notification engine - Push notification service integration (FCM, APNs) - SMS integration (Twilio/similar) - Email service integration (SendGrid/similar) - Notification template management - Communication log/history	ADMIN PORTAL BACKEND APIs - Admin user management - RBAC for 7 distinct admin roles - Customer management APIs - Service request management APIs (admin operations) - Contractor configuration APIs - Manual processing queues - Failed enrollment queue - Override/exception handling APIs	DATA SYNC & INTEGRATION LAYER - Commerce CRM integration (Duke customers) - Dynamics integration (P&G customers) - Customer profile sync logic - HPP plan sync - Service request bi-directional sync - Contractor data sync from CRM - Sync error handling & retry logic
ANALYTICS & REPORTING BACKEND - Analytics instrumentation - Event tracking system - KPI calculation logic - Dashboard data aggregation APIs - Report generation & export functionality		

📌 Backend represents 42% of development investment due to complex contractor matching, dual service flows (HPP + ad-hoc), multi-system data sync, and comprehensive admin capabilities.

DevOps, Deployment & Security Breakdown

Infrastructure, Compliance & Operations (\$171,292)

INFRASTRUCTURE SETUP & CONFIGURATION

- Cloud infrastructure setup (AWS/Azure/GCP)
- Environment provisioning (dev, staging, production)
- Load balancing & auto-scaling configuration
- CDN configuration
- DNS management
- SSL/TLS certificate management
- Backup strategy implementation

CI/CD PIPELINE

- Git repository setup & branching strategy
- CI/CD pipeline configuration (GitHub Actions/GitLab CI/Jenkins)
- Automated testing integration
- Build & deployment automation
- Environment-specific configuration management
- Rollback procedures

DATABASE & STORAGE

- Database provisioning (MySQL/PostgreSQL)
- Database replication setup
- Backup & recovery automation
- S3/cloud storage setup (images, PDFs, documents)
- Database migration strategy
- Data retention policies
- Database performance optimization

SECURITY IMPLEMENTATION

- Duke Energy cybersecurity requirements compliance
- Role-based access control implementation
- Multi-factor authentication (MFA)
- API security (rate limiting, JWT validation)
- Data encryption at rest & in transit
- Security headers configuration
- OWASP Top 10 vulnerability mitigation
- Penetration testing preparation
- Security audit documentation
- PCI compliance prep (for future payment integration)

MONITORING & LOGGING

- Application monitoring setup (New Relic/Datadog/CloudWatch)
- Log aggregation (ELK stack/CloudWatch Logs)
- Error tracking (Sentry/Rollbar)
- Performance monitoring
- Uptime monitoring & alerting
- Dashboard creation for DevOps metrics
- Alert configuration & on-call setup

API GATEWAY & MANAGEMENT

- API gateway setup
- API versioning strategy
- API documentation hosting
- API rate limiting & throttling
- API key management
- API analytics

CONTAINER & ORCHESTRATION

- Docker containerization
- Kubernetes/ECS orchestration setup
- Container registry setup
- Service mesh configuration (if needed)
- Container security scanning

NETWORK & VPN CONFIGURATION

- VPC setup
- Security group configuration
- Network ACLs
- VPN setup for Duke Enterprise API access
- Private subnet configuration for databases

COMPLIANCE & DOCUMENTATION

- Compliance documentation (Duke Energy requirements)
- Architecture documentation
- Runbook creation
- Disaster recovery planning
- Data privacy controls documentation
- Security policy documentation

MOBILE APP DEPLOYMENT

- iOS App Store submission & review
- Android Play Store submission & review
- App signing certificate management
- App update strategy
- Push notification certificate setup
- Mobile analytics configuration

📌 DevOps represents 21.7% of investment, critical for Duke Energy cybersecurity compliance, enterprise-grade security standards, and multi-environment infrastructure supporting 200K-2M users.

Next Steps & Discussion

01

Proceed Through Legal

Finalize contract terms, review updated scope and budget, execute agreements and establish project governance framework

02

Team Building / Site Visit

Introduce core development team members, conduct site visit to understand Duke Energy environment, establish communication channels and working relationships

03

Agree on Project Approach

Confirm project scope and timeline, align on bridge functionality strategy, establish FSM/API integration milestones and transition plan

04

Project Kick-off

Formal project launch meeting, establish sprint cadence and ceremonies, set up development environments, confirm technical architecture decisions

05

Proceed with Development Planning

Detailed sprint planning, user story refinement, technical design documentation, establish QA and testing protocols, begin development work

Important Considerations

- This is a realistic estimate with contingency, not a final fixed price
- Actual costs will vary based on business decisions and third-party tool timing
- Many variables remain around integration dependencies
- Phasing decisions will impact timeline and resource allocation

Ready for Questions

We're prepared to discuss any aspect of the discovery findings, scope changes, phasing approach, or budget considerations.

Let's align on priorities and chart the path forward together.